companies and for thinly-traded securities.[1229] This is because narrower quoted spreads would discourage some liquidity providers from entering the market. The Commission disagrees with this characterization. The academic research on the Tick Size Pilot (TSP), which increased the tick size for some smaller stocks from $0.01 to $0.05 between 2016 and 2018, suggests that for many stocks affected by the TSP, particularly those with narrower quoted spreads, the TSP led to worse market quality.[1230]

Additionally, lowering excess rents and the oversupply of liquidity caused by tick size induced pricing distortions is likely to reduce aggregate depth across all price levels. However, this reduction is unlikely to be harmful to overall market quality, even for smaller or thinly traded securities, as it would relieve a distortion resulting in an oversupply of liquidity. As the amount of liquidity provision comes closer to equilibrium levels, quoted spreads narrow and queue lengths shorten, lowering transaction costs and increasing the likelihood that relatively slower fundamental and/or retail traders could interact with each other. This will reduce total transaction costs for these traders because one side would be earning the quoted spread on the transaction.[1231]

ii. Empirical Analysis

This section presents the Commission's empirical analysis, as well as a discussion of commenter analysis and views concerning the effect of a tick size reduction on various aspects of market quality. Based on these analyses, the Commission concludes that on average, stocks that receive the smaller $0.005 tick size will experience improved market quality – implying that, for

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1229 See STA Letter at 5.

1230 See infra section VII.D.1.b.ii and Barardehi et al., supra note 231 for additional discussion of the tick size literature.

1231 See Retirement Coalition Letter at 2, Pragma Letter at 10.

these affected stocks, the predominant market quality effect of the smaller tick size will be an increase in pricing efficiency.[1232]

The academic literature examining the effect of tick sizes on financial markets largely studies two events: decimalization, which occurred in 2001[1233] and reduced the tick from 1/16th of a dollar ($0.0625) to $0.01; and the TSP, which ran from October 2016 to October 2018 and temporarily increased the minimum tick increment from $0.01 to $0.05 for a sample of small cap stocks.[1234] Most of the literature surrounding decimalization suggests that, on average, decimalization was associated with a decline in quoted spreads consistent with the notion that lowering the tick size relieved distortions related to having a tick size that is too wide.[1235]

In the Proposing Release, the Commission supplemented existing research with its own analysis on the TSP.[1236] As stated in the Proposing Release, market dynamics have changed dramatically in the more than two decades since decimalization. Most notably over that period, electronic, algorithmic, and high-frequency trading have come to dominate the trading landscape,

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1232 See supra note 1199 and surrounding text for additional discussion of the tick size tradeoff.

1233 See, e.g., Order Directing the Exchanges and the National Association of Securities Dealers, Inc. to Submit a Phase-in Plan to Implement Decimal Pricing in Equity Securities and Options; Pursuant to Section 11A(a)(3)(B) of the Securities Exchange Act of 1934, Securities Exchange Act Release No. 42914 (June 8, 2000), 65 FR 38010 (June 19, 2000); Commission Notice: Decimals Implementation Plan for the Equities and Options Markets, SEC (July 24, 2000), available at http://www.sec.gov/rules/other/decimalp.htm.

1234 See Edwin Hu, et al. (2018), supra note 1002, for additional details about the Tick Size Pilot.

1235 See Hendrick Bessembinder, Trade Execution Costs and Market Quality After Decimalization, 38. J. FIN. & QUANTITATIVE ANALYSIS 747 (2003). See also Michael A. Goldstein & Kenneth A. Kavajecz, Eighths, Sixteenths and Market Depth: Changes in Tick Size and Liquidity Provision on the NYSE, 56 J. FIN. ECON. 125 (2000) and Charles M. Jones & Marc L. Lipson, Sixteenths: Direct Evidence on Institutional Execution Costs, 59 J. FIN. ECON. 253 (2001), both examining the earlier tick size change from 1/8 to 1/16 of a dollar. See also Sugato Chakravarty, Venkatesh Panchapagesan & Robert A. Wood, Did Decimalization Hurt Institutional Investors?, 8 J. FIN. MKTS. 400 (Nov. 2005) and Sugato Chakravarty, Bonnie F. Van Ness, & Robert A. Van Ness, The Effect of Decimalization on Trade Size and Adverse Selection Costs, 32 J. BUS. FIN. & ACC. 1063 (June/July 2005), both suggesting that large institutional trades may have become more costly following decimalization.

1236 See Proposing Release, supra note 11, at 80318-80322.

whereas they were much less prominent in 2001. These changes diminish the relevance of evidence from these prior periods, making it desirable to supplement existing studies with evidence that is closer in time [1237]

Some commenters questioned using the TSP to estimate the effects of a reduced minimum pricing impact because the TSP affected only a subset of small cap stocks, did not contain ETPs, and did not affect access fee caps.[1238] One of those commenters suggested that the TSP analysis was not applicable because it focused on stocks with quoted spreads much wider than the few cent quoted spreads contemplated by the amendments.[1239] The same commenter suggested that the TSP was not applicable because it applied to a 5 to 1 tick size change, which is different from the tick size change in the amendments.[1240] Some commenters went further and questioned whether anything could be learned from the TSP because it did not involve subpenny tick sizes.[1241]

As explained in the following discussion, the Commission continues to believe that the TSP provides a meaningful environment to study the potential effects of a tick size change for the reasons articulated below, even as the TSP has limitations for determining the exact effect of the amendments to Rule 612.

1238 See, e.g., CCMR Letter at 27, Virtu Letter II at 64. Lewis Letter attached to Virtu Letter II at 34.

1239 See CCMR Letter at 27.

1240 Id.

1241 See Citadel Letter I at 12 (stating that the TSP "provides no information on what would be expected to occur if minimum quoting increments were further reduced to levels that have never before been tested"); and Virtu Letter II at 3 ("The TSP studied the impact of a widened minimum quoting and trading increment for certain small capitalization stocks, and offered no analysis, data, or conclusions on the potential impact that a narrowed, sub-penny tick regime would have on the marketplace, the investor experience, or issuers. It is an apples-to-oranges comparison and is irrelevant as a basis for support").

First, the economics of being tick-constrained do not depend on the absolute size of the tick in question. Rather, they depend on the relationship between the economic spread[1242] implied by the economics of the stock and the quoted spread that is possible given the tick size, regardless of the specific tick size. Specifically, when the economic spread is narrower than a single tick, the negative effects of being tick-constrained are expected to emerge for the reasons discussed in section VII.D.1.b.i above.[1243] Those reasons are independent of the absolute size of the tick. They instead depend on the ratio of the market price of liquidity to the tick, i.e., the lowest quoted spread permitted by the tick size. In this context, the TSP analysis has merit, even as it includes some stocks with quoted spreads wider than 1.5 cents (the cutoff for the amendments), because its purpose is to gain insight into how stocks with various numbers of tick increments intra-spread react to changing the tick. Addressing this question necessitates considering stocks with wider quoted spreads.

Second, as discussed above, a key factor in the economics of being tick constrained is the implied quoted spread relative to the tick size, not the market capitalization or any other the qualifying factors for the TSP. Because the economics of being tick-constrained do not depend on market capitalization, the findings derived from the TSP can be usefully applied to a broader section of the market. Also, one study,[1244] referenced in the Proposing Release, specifically examined only the most liquid TSP stocks and removed stocks with very low prices.[1245] The authors' results indicate that, in this subset of the most liquid TSP stocks, all key findings of the

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1242 See supra note 990 and surrounding text for discussion of the term economic spread.

1243 See also id. for a discussion of the concept of economic spread.

1244 See Barardehi et al., supra note 231.

1245 See Proposing Release, supra note 11, at 80273 n.85.

TSP not only hold, but that the patterns of the results of the TSP on market outcomes tend to strengthen.

Third, while the Commission acknowledges the difference between the TSP and the amendments with regard to tick size splits – the TSP was a 1:5 tick size change while the amendments provide a 1:2 tick size change for some stocks – the TSP provides meaningful information about the likely direction of the effects due to a tick size change: that is, whether market quality improves or declines when the tick size is changed. The actual effect of a 1:2 split may differ from that observed from the TSP's 1:5 split, but it is unlikely to go in the opposite direction if the TSP were to indicate a market quality improvement when the tick size is reduced. This is because the potential negative effects of too many ticks intra-spread would be stronger for the 1:5 split associated with the TSP than with the 1:2 split associated with the amendments.[1246] Thus, it is unlikely that, were the TSP to show an improvement in market quality associated with a 1:5 tick size split for certain stocks, that there would have been an opposite effect with a 1:2 tick size split.

Some commenters stated that reducing the tick size below $0.01 was opposed to the conclusions and analysis provided by the Commission when adopting Rule 612 in 2005, and that the Commission did not provide analysis explaining why it was reversing its opinion.[1247] The Commission disagrees that the analysis and conclusions associated with the initial proposal and adoption of Rule 612 are inconsistent with the analysis provided in the Proposing Release and

1246 For example, the potential negative effects from sub-pennying would be higher from a 1:5 split compared to a 1:2 splits because the cost of gaining priority over other liquidity providers, by updating the best price by a single tick, is lower with a smaller tick. The risk liquidity fragmenting across price levels would also be higher with a 1:5 split as compared to a 1:2 split. See supra section VII.D.1.b.i for further discussion.

1247 See, e.g., Equity Market Structure Citadel Letter at 17, Craig Louis Letter attached to Virtu Letter II at 32-33, and Virtu Letter II at 16. See supra section VII.C.1.b for a discussion of the Commission analysis referred to by commenters.

repeated here. When initially proposing and adopting Rule 612, the Commission acknowledged that lowering the tick size from fractions to $0.01 improved the trading environment.[1248] It also expressed concern, as stated by commenters, that further reducing the tick size for all stocks could harm market quality via pennying and reduced liquidity at the top of the book.[1249] When originally proposing Rule 612, the Commission also provided an analysis of sub-penny trading and quoting and stated that there was, at the time, no industry standard for trading and quoting increments.[1250] The Commission's sub-penny analysis suggested that, at the time, sub-penny trading was primarily used to facilitate pennying because sub-penny trades congregated at $0.001 and $0.009 rather than having a uniform distribution or clustering midpoint prices (i.e., in $0.005 increments), justifying the use of some minimum pricing increment.[1251]

When adopting Rule 612, the Commission did not empirically analyze whether a minimum pricing increment of $0.005 would have harmed or helped market quality for some stocks, and specifically did not opine on a tiered tick structure such as is being adopted. The analysis provided therein was in the context of a uniform tick size applicable to all stocks. Within this context, the Commission concluded that "the marginal benefits of a further reduction

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1248 See 2004 Regulation NMS Proposing Release, supra note 31, at 11170.

1249 Id.; see also Proposing Release, supra note 11, at 80280 ("Minimum pricing increments that are too small can also add to complexity in trading and increase the risk of stepping ahead"); see also supra note 994 for the definition and discussion of pennying.

1250 Id. at 11171. Although Nasdaq and the exchanges permitted quoting in single penny increments, these markets allowed trades to be printed in increments below a penny. Although certain online brokers only accepted orders priced in one-cent increments, ECNs and Nasdaq market makers accepted orders and executed trades in sub-penny increments. While market makers quoted through Nasdaq only in penny increments, they could display orders in ECNs in sub-pennies. Exchanges, where the majority of trading volume occurred, were bound by the Decimals Implementation Plan, which was ordered by the Commission, and which ultimately established $0.01 as the tick size for exchange quotes. Other market participants, however, were not so bound leading to non-standard quoting increments across various venues such as ECNs.

1251 Id. at 11169.

in the minimum pricing increment [below $0.01 for all stocks] are not likely to justify the costs to be incurred by such a move"[1252] The analysis provided in this release does not disagree with that assessment. Applying a tick size lower than $0.01 for all stocks could cause harm to stocks with wide quoted spreads due to pennying concerns and fragmenting liquidity.

Additionally, the need to address tick-constrained stocks has increased substantially in the subsequent nearly two decades as tick constraints have become more pervasive over time. Table 3 indicates that in 2023, about 74% of share volume was associated with securities trading with quoted spreads at or below $0.015; following the same methodology, in 2005 the figure was about 54%.[1253] This statistic understates the true increase in trading in tick-constrained securities because overall average daily trading volume has more than doubled over the same period of time.[1254] Thus, precisely because average quoted spreads have been coming down, the benefits of alleviating the tick constraint have increased substantially since 2005. Additionally, as discussed throughout this section, there has been considerable research since the implementation of Rule 612 in 2005 by the Commission, industry members, and academics surrounding tick sizes that did not exist when Rule 612 was adopted. This research supports the notion that a tick size below $0.01 will likely improve market quality for some stocks.

One commenter illustrated its disagreement with the Commission's use of the TSP analysis by presenting a hypothetical TSP in which the tick size is increased from $0.01 to

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1252 Id. at 11170.

1253 These patterns are not driven by a change in sub-dollar trading (which may benefit from a narrower tick size); the patterns are not materially changed when symbol-days with average prices below $2 or $5 are dropped from the sample.

1254 This statistic is computed by comparing the average daily share volume in all securities covered by WRDS Intra-day indicators in 2005 and 2023. Additionally, total trading volume has also more than doubled over that same time period. Thus, there is more trading volume and more of it is trading in a tick-constrained environment.

$0.15.[1255] The commenter stated that such a change "would have negatively impacted a greater range of stocks … and predictably liquidity conditions in those stocks would have meaningfully improved at the end of the pilot when the changes were reversed."[1256] The commenter proceeded to state that "this experiment would not suggest that regulators should always reduce the minimum quoting increment for tick-constrained symbols by a factor of *fifteen*."[1257] The commenter further stated that the TSP "merely reverted to the *status quo* after a failed experiment", and this "reversion provides no information on what would be expected to occur if minimum quoting increments were further reduced to levels that have never before been tested."[1258]

The Commission disagrees with this assessment in several respects and continues to believe that analysis of the TSP provides meaningful information for the effects of the amendments. The TSP enables analysis that empirically tests whether market quality depends on being tick-constrained. The TSP provides two events that can be used for this test: one at the start of the TSP when tick sizes were increased for certain stocks, and one at the end of the TSP where tick sizes for those same stocks were decreased. Academic research shows that the effects of both of these events are consistent with the theory that stocks with few ticks intra-spread have worse market quality.[1259] The Commission provided its own analysis of the end of the TSP; the end of the TSP involved a reduction in tick size, which directionally corresponds to what will happen under the adopted rule. This analysis found evidence to support the theory that stocks

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1255 See Citadel Letter I at 12.

1256 Id.

1257 Id. (emphasis in original).

1258 Id.

1259 See, e.g., Barardehi et al., supra note 231.

with too few ticks intra-spread have worse market quality. The Commission therefore disagrees that the TSP analysis "provides no information" as to the effects of the adopted rules.

The commenter states that "the end of the Tick Size Pilot provides no basis for suggesting that regulators should always reduce the minimum quoting increment for tick-constrained symbols by a factor of five." The Commission does not reach the conclusion that regulators should always reduce the minimum quoting increment by a factor of five, and indeed the Commission is not adopting such a rule. As discussed earlier in this section, TSP analysis indicates that for stocks with 1-2 ticks intra-spread, reducing the tick size improves market quality on average. While the Commission is reducing the tick size by a factor of two rather than five for some stocks, the direction is likely to be the same as what was observed in the TSP, though the magnitude may be different.[1260]

Furthermore, the commenter assumes in the hypothetical experiment of a bigger increase in the tick size, that this increase would have "negatively impacted" stocks. However, the fact that causing stocks to become tick-constrained worsens their market quality is an assumption made by the commenter. Absent evidence, such as the evidence provided by the TSP, it is unclear upon what the commenter bases this assumption. The ability to make this inference, that being tick-constrained worsens market quality, is precisely why analyzing the TSP is valuable because it provides the empirical result which permits one to employ with confidence the commenter's assumption in its hypothetical.[1261]

1260 See supra note 1246 and surrounding text for additional discussion.

1261 More specifically, the commenter's hypothetical assumes that the start of a large tick size increase would worsen liquidity, then concludes this means that evidence from the end of the TSP is uninformative because it simply reverses the effect. But, this conclusion is incorrect because the TSP results from both the imposition and conclusion of the TSP are what make the first assumption credible.

With regard to the commenter's statement that the TSP conclusion "provides no information on what would be expected to occur if minimum quoting increments were further reduced to levels that have never before been tested,"[1262] while it is true that the tick sizes in the adopted amendments are not among the tick sizes implemented in the TSP, this fact does not render the TSP analysis uninformative. The theory that stocks that are tick-constrained will trade better with more ticks intra-spread, successful as it was in predicting the market quality effects of the end of the TSP, can be reasonably relied upon to help determine the effects of the amendments. What matters is not the magnitude of the spread, or the size of the tick, but the number of ticks intra-spread.[1263]

One commenter stated that the analysis in the Proposing Release should have accounted for both fixed effects and volatility, as well as other variables.[1264] Barardehi et al. (2022) provide estimates that account for fixed effects and a number of control variables including volatility.[1265] This paper shows that the results shown in the Proposing Release and repeated below are robust to these effects.

Commission Empirical Analysis: The Proposing Release provided a review of the existing TSP empirical academic research.[1266] This research consistently found that stocks that became tick-constrained by the TSP, on average, traded better across many market quality dimensions when their tick size was reduced from $0.05 to $0.01. Some analysis also showed

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1262 See Citadel Letter I at 12.

1263 See infra section VII.F.1 for a discussion of reasonable alternative tick sizes.

1264 See Virtu Letter II at 15.

1265 See Barardehi et al., supra note 231. The authors control for market capitalization, dollar volume, average quoted spread, and return volatility, see analysis associated with their Table 11.

1266 See Proposing Release, supra note 11, section V.D.1.